NETFLIX

DATA ANALYSIS: INSIGHTS & TRENDS

DATASET OVERVIEW

1. DATA SOURCE :

- Provider: Kaggle (Open Data Platform).
- Dataset: Netflix Movies and TV Shows.
- Scope: Comprehensive data on titles available on the platform up to 2021.

2. DATASET DIMENSIONS :

- Total Entries: 8,807 rows (unique titles)
- Total Features: 12 columns (key attributes).

3. KEY ATTRIBUTES FOR ANALYSIS :

- Type: Categorizing content into Movies vs. TV Shows.
- Country: Analyzing global production hubs
- Rating: Understanding age-based target audiences.
- Date_added: Investigating temporal trends and seasonality.

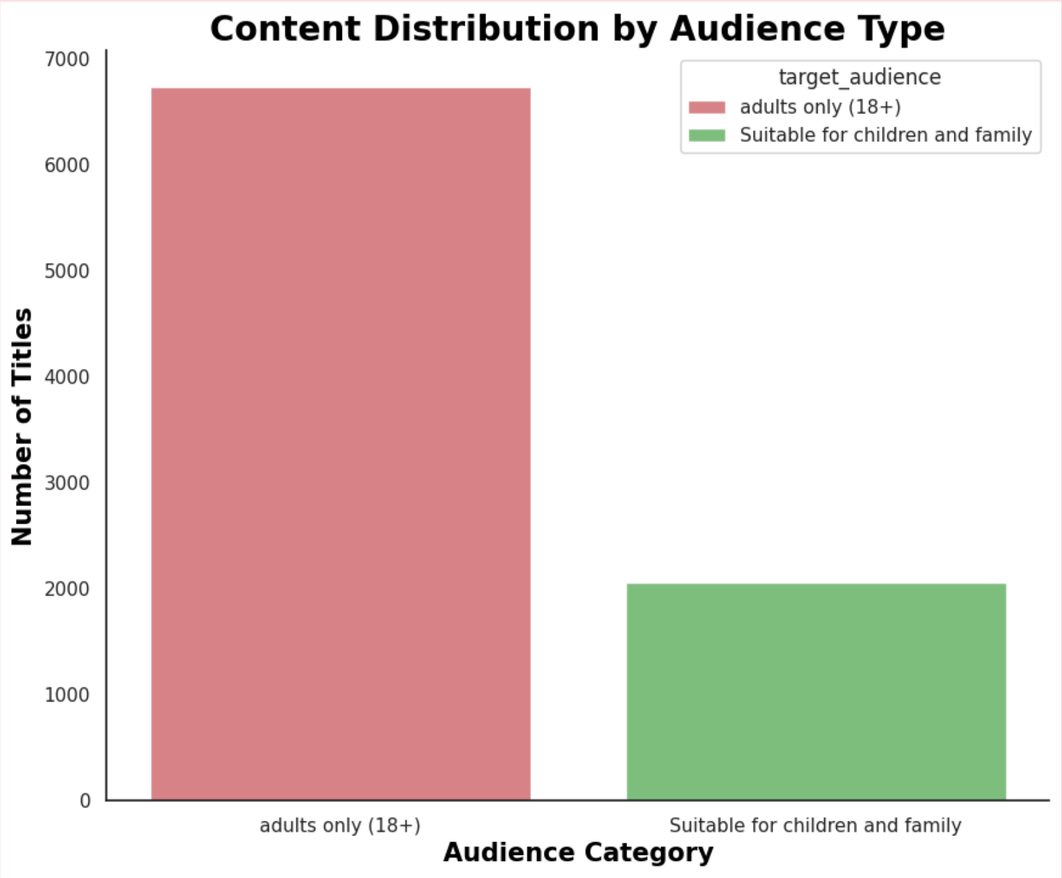
4 DATA INTEGRITY :

- Cleaning: Handled missing values (NaN) in columns like Director, Cast, and Country.
- Processing: Standardized date formats to enable accurate time-series visualization.

TARGET AUDIENCE DISTRIBUTION

KEY INSIGHTS :

- Top Category (TV-MA): The clear leader in content volume, indicating that Netflix's largest library segment is designed for Mature Audiences (18+)
- Secondary Tier (TV-14 & R): Significant content for Teens (14+) and restricted adult viewing, showing a broad but age-specific strategy
- Minority Categories (TV-G, TV-Y7, PG): These categories have lower counts, showing that Netflix is less focused on children’s content compared to its adult catalog



Netflix positions itself as a "Premium Adult Streaming Service," where the majority of its investment goes into complex, mature storytellin

Step-by-Step Process

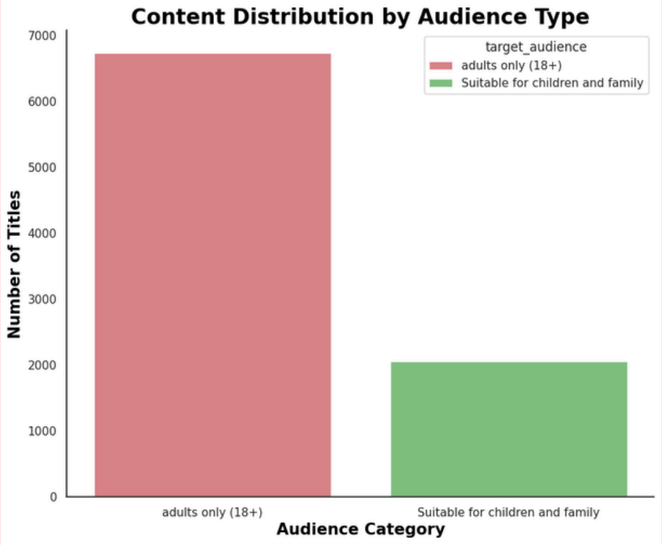
- Frame Setup: Initialized the plot dimensions using plt.figure(figsize=(10,8)) for clear visibility.

- Visual Mapping: Used sns.histplot to represent the distribution of the target_audience column

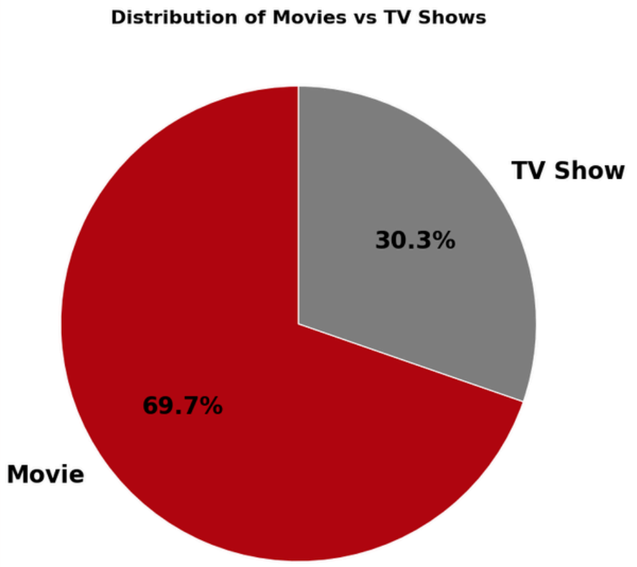
- Color Styling: Applied a custom palette (#B20710 for Netflix Red and Green) to differentiate between audience categories.

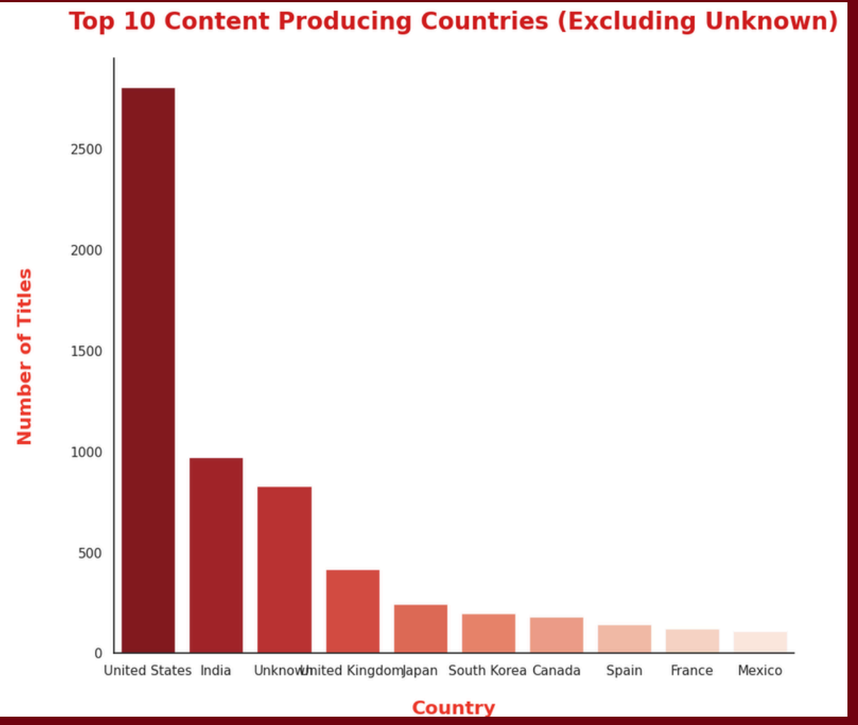
- Refinement: Used shrink=.8 and sns.despine() to improve the chart's readability and remove unnecessary borders.

- Labeling: Enhanced the chart with bold titles and axis labels for professional presentation.



NETFLIX CONTENT STRATEGY & GLOBAL PRESENCE :





NETFLIX USES A "HIGH-VOLUME MOVIE" STRATEGY, PRIMARILY DRIVEN BY THE US MARKET BUT HEAVILY SUPPORTED BY A DIVERSE GLOBAL PRODUCTION NETWORK.

PART A :

CONTENT COMPOSITION MOVIES VS. TV SHOWS

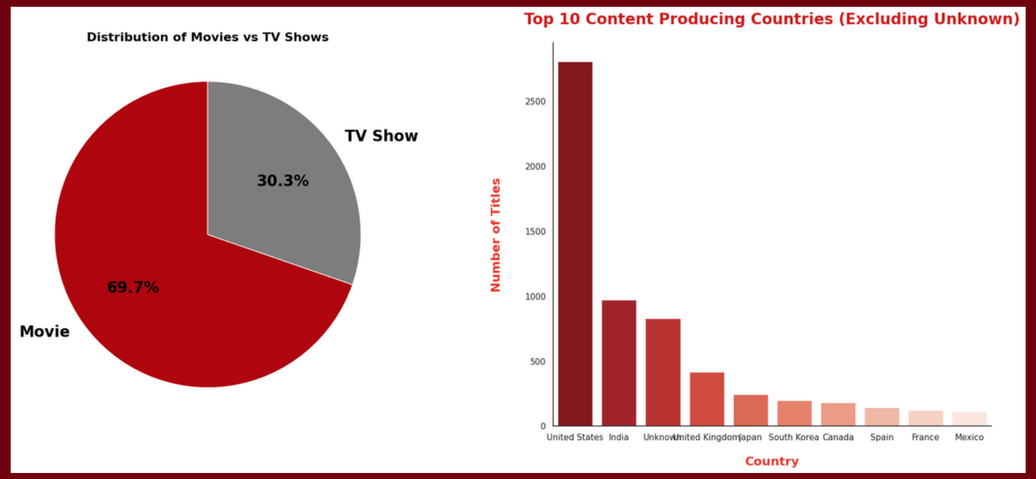
- DOMINANCE: MOVIES LEAD WITH NEARLY 70% OF THE TOTAL LIBRARY
- VOLUME: OVER 6,100 MOVIES COMPARED TO APPROXIMATELY 2,600 TV SHOWS

PART B :

GEOGRAPHICAL LEADERS TOP 10 COUNTRIES

- TOP PRODUCER: THE UNITED STATES IS THE PRIMARY CONTRIBUTOR BY A SIGNIFICANT MARGIN.
- GLOBAL HUBS: INDIA AND THE UNITED KINGDOM FOLLOW, HIGHLIGHTING A STRONG PRESENCE OF BOTH HOLLYWOOD AND BOLLYWOOD CONTENT.
- DIVERSIFICATION: THE LIST INCLUDES COUNTRIES LIKE SOUTH KOREA, SPAIN, AND CANADA, SHOWING NETFLIX'S COMMITMENT TO INTERNATIONAL STORYTELLING

Step-by-Step Process



- Subplot Configuration: Used plt.subplot(1, 2, 1) and (1, 2, 2) to create a side-by-side layout, allowing for a direct comparison between content types and geographical data.

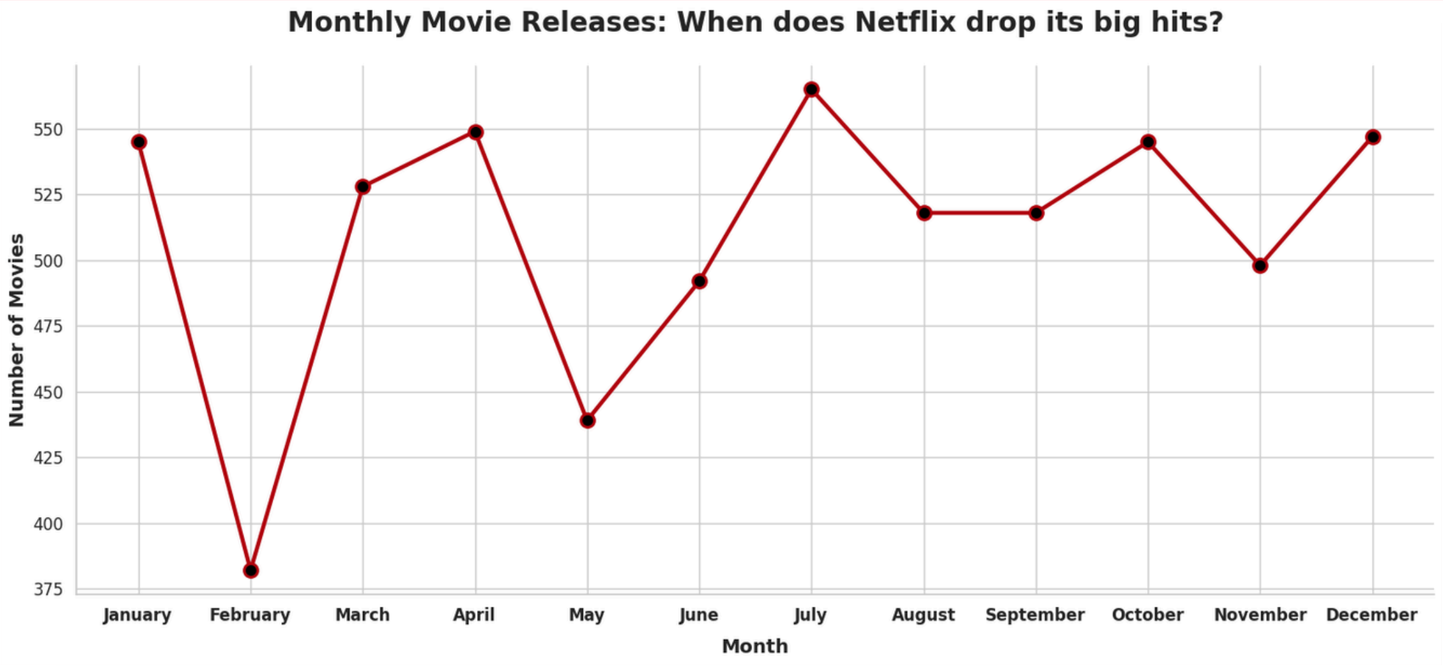
- Data Aggregation: * Applied .value_counts() on the type column to determine the ratio of Movies vs. TV Shows.

- Extracted the top 10 producing nations using .value_counts().head(10) from the country column.

Visual Styling:

- Left (Pie Chart): Implemented a custom Netflix-themed palette (#B20710 and Gray) with autopct to show precise percentages.
- Right (Bar Chart): Used a gradient palette (Reds_r) to rank countries, enhancing the visual hierarchy of the top producers.
- Layout Optimization: Applied plt.tight_layout(pad=5.0) and sns.despine() to ensure the charts don't overlap and look clean

MONTHLY RELEASE TRENDS

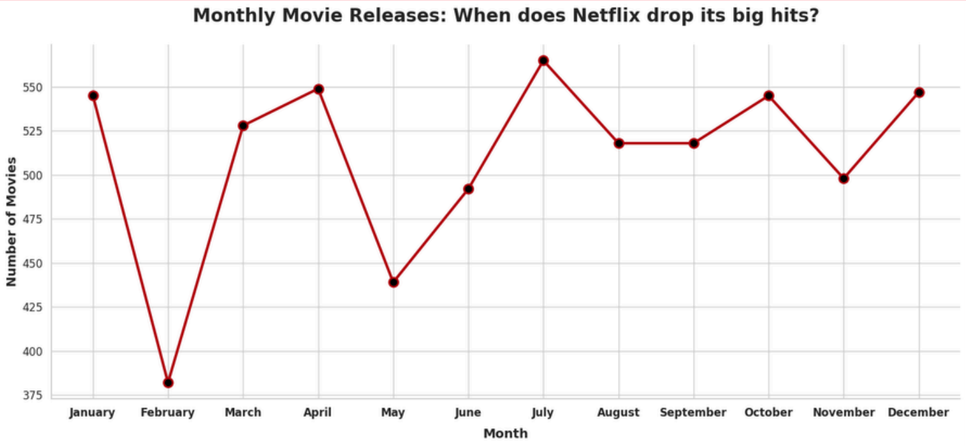


SUCCESSFULLY PERFORMED TIME-SERIES DATA CLEANING BY CONVERTING RAW TEXT INTO STANDARD DATE FORMATS TO EXTRACT THESE MONTHLY PATTERNS.

KEY INSIGHTS :

- RELEASE PEAKS: A NOTICEABLE INCREASE IN MOVIE RELEASES DURING SPECIFIC MONTHS (E.G., JULY, DECEMBER, OR JANUARY).
- STRATEGIC TIMING: NETFLIX OFTEN SCHEDULES MAJOR MOVIE DROPS TO COINCIDE WITH GLOBAL HOLIDAY SEASONS AND SCHOOL BREAKS.
- CONSISTENCY: WHILE RELEASES HAPPEN YEAR-ROUND, THE DATA SHOWS "SPIKES" THAT MATCH PEAK VIEWER AVAILABILITY.

Step-by-Step Process



Data Cleaning & Preprocessing:

- String Sanitization: Used .str.strip() to remove any hidden white spaces in the date_added column that might cause errors during conversion.
- Datetime Conversion: Converted the raw text dates into official Datetime objects using pd.to_datetime.

Feature Engineering:

- Month Extraction: Used .dt.month_name() to create a new feature (month_added) to identify exactly which month each title was released.
- Logical Sorting: Defined a custom months_order list to ensure the chart follows the calendar year (January to December) instead of alphabetical order.

Visual Design:

- Plotting: Created a Line Plot using a bold Netflix-red color (#B20710) with black markers to highlight specific data points.
- Customization: Adjusted the linewidth and markersize for better visibility during the presentation

شكرا "جسور" تحدي 30 يوم

NAME : ALANOUD SAUD

Email : almtyryalnwd870@gmail.com

MOBILE NUMBER : 0505901490